



VocalIQ · India Language Roadmap

Har state, har bhasha — India-first AI voice agents.

Hindi + 22 regional languages ke liye best models (Sarvam AI, AI4Bharat, Bhashini) ka integration roadmap, cost comparison, 22-language coverage map, aur "go-massive across India" strategy. **VocalIQ ko India ka #1 vernacular voice-AI platform banane ka plan.** Research: July 2026 (200+ official sources).

Ek nazar mein

22

INDIAN LANGUAGES
(SARVAM+AI4BHARAT)

~\$0.02–0.03

INDIA CALL / MIN (SARVAM)

83x

TTS SASTA VS ELEVENLABS

3 din

SARVAM ADAPTERS (PHASE 1)

 **Abhi ka gap:** VocalIQ ka default **Deepgram (STT) Hindi ke liye TOOTA hua hai** (2 independent benchmarks confirm), aur ElevenLabs ki Hindi "robotic" — Bengali/Kannada/Gujarati weak. Matlab India market me current setup kamzor. **Achhi baat:** aapka provider-router architecture perfect hai — Indian models bas **naye adapters** hain, koi rewrite nahi.

1

Winner: Sarvam AI — VocallIQ ke liye perfect fit

India ki "sovereign AI" — poori Indic stack (STT+TTS+LLM+translation), aapke exact tech-stack (Pipecat+LiveKit) me native. Apache 2.0 + API.

PRODUCT	KYA	LANGUAGES	PRICE	FIT FOR VOCALIQ
Saaras v3	STT	22 Indian + code-mixed Hinglish	₹30/hr ≈ \$0.006/min	WebSocket streaming, mulaw@8kHz telephony, auto-detect. Hindi WER ~5-6%.
Bulbul v3	TTS	11 Indian (39 voices)	₹30/10K ≈ \$3.6/1M chars	Sub-250ms, natural Indian prosody. ElevenLabs se 83× sasta.
Sarvam-30B / 105B	LLM	22 Indian (custom tokenizer)	\$0.03/\$0.12 · \$0.048/\$0.19 /1M	OpenAI-compatible (api.sarvam.ai/v1) → router me drop. GPT-4o se ~100× sasta.
Mayura / Sarvam-Translate	Translate	11 / 23 Indian	₹20/10K ≈ \$0.024/10K	Code-mixed Hinglish style, script control.

✔ **Kyun perfect:** (1) **Pipecat + LiveKit dono me native Sarvam connectors** — aapka voice service directly plug. (2) LLM OpenAI-compatible — 5 min me provider-router me. (3) **Apache 2.0 open weights** — self-host bhi (CodeCanyon/BYOK). (4) Free ₹1,000 credits + startup program (6–12 mo free). (5) mulaw@8kHz + WebSocket = phone-call ready. (6) Code-mixed Hinglish first-class — Indians aise hi bolte hain.

⚠ *Caveats: Bulbul TTS abhi 11 languages (baaki 12 ke liye AI4Bharat/Cartesia fallback); English LLM quality weak (English ke liye GPT/Claude rakho); early-stage company → multi-provider routing me layer karo (sole provider mat banao). 250ms latency Sarvam ka apna claim — khud benchmark karo.*

2

Recommended India stack (hybrid — router chooses)

LAYER	PRIMARY (INDIA)	FALLBACK	KYUN
STT	Sarvam Saaras v3 (22 lang)	Deepgram (English only)	Best Hindi/regional, \$0.006/min, streaming.
LLM	Sarvam-30B / Gemini Flash	Claude/GPT (English)	Indic reasoning, sasta, OpenAI-compatible.
TTS	Sarvam Bulbul v3 / Cartesia (9 Indic, 40ms)	ElevenLabs	Natural Indian voice, 40ms latency.
Translation	Sarvam Mayura / IndicTrans2	LLM translate	Code-mixed Hinglish.
Self-host	AI4Bharat (IndicConformer + IndicF5 + IndicTrans2)	—	MIT, free — near-zero cost at scale, BYOK/CodeCanyon.
Govt/free	Bhashini (22 lang, subsidized)	—	Prototype + government/PSU clients.

3

Provider comparison — Indian languages ke liye

PROVIDER	INDIAN LANGS	REAL-TIME	COST	HINDI QUALITY	VERDICT
Sarvam 🇮🇳	STT 22 · TTS 11	Yes (WS)	\$0.006/min STT	★★★★★	Primary — best fit.
AI4Bharat 🇮🇳 (self-host)	22 (MIT)	RNNT streaming	GPU only ~\$0.005/min	★★★★☆	Self-host / BYOK / scale.
Bhashini 🇮🇳 (govt)	22	Variable	Free/subsidized	★★★☆☆	Prototype + govt (no SLA).
Gnani.ai 🇮🇳	12+ (telephony-trained)	Yes <500ms	Enterprise quote	★★★★★	BFSI contact-center partner.
Cartesia	TTS 9 Indian	40ms TTFA	~\$33/1M chars	★★★★☆	Best mainstream Indian TTS.
Deepgram (current)	6 (Hindi broken!)	Yes	\$0.0077/min	★☆☆☆☆	English only — Hindi mat use.
ElevenLabs (current)	3-4 (robotic)	Yes	~\$300/1M chars	★★★☆☆	English/premium only.
Google / Azure	10-12	Yes	~\$0.016/min	★★★☆☆	22-30% WER — fallback only.

Source: Voice-of-India benchmark 2026 + 8-provider STT benchmark. Sarvam #1 in 13/15 languages; Deepgram Nova-3 Hindi detection confirmed broken by 2 independent sources.

4

22-Language Coverage Map — kaun se state, kaun sa stack

✔ = end-to-end ready (STT+TTS) · 🕒 = STT ready, TTS via AI4Bharat/Cartesia fallback. India ke ~95% bolne wale coverage me.

LANGUAGE	MAJOR STATES	SARVAM STT	SARVAM TTS	AI4BHARAT	E2E READY
Hindi	UP, MP, Bihar, RJ, Delhi, HR	✔	✔	✔	✔
Bengali	West Bengal, Tripura	✔	✔	✔	✔
Marathi	Maharashtra	✔	✔	✔	✔
Telugu	Andhra Pradesh, Telangana	✔	✔	✔	✔
Tamil	Tamil Nadu, Puducherry	✔	✔	✔	✔
Gujarati	Gujarat	✔	✔	✔	✔
Kannada	Karnataka	✔	✔	✔	✔
Malayalam	Kerala	✔	✔	✔	✔
Punjabi	Punjab	✔	✔	✔	✔
Odia	Odisha	✔	✔	✔	✔
Urdu, Assamese	J&K, Assam	✔	🕒	✔	🕒
+11 more (Sanskrit, Kashmiri, Sindhi, Nepali, Konkani, Manipuri, Bodo, Dogri, Maithili, Santali)	NE, hills, regional	✔	—	✔	🕒 (AI4Bharat TTS)

🇮🇳 **India domination:** Top-11 languages (India ki ~95% population) **end-to-end ready** Sarvam se. Baaki 11 STT + AI4Bharat/Bhashini TTS se cover. **Koi global competitor itni breadth nahi deta** — yahi VocalIQ ka India moat hai.

5

Cost comparison — India call ka per-minute

STACK	STT	LLM	TTS	TOTAL/MIN	HINDI QUALITY
CURRENT (Deepgram+ElevenLabs+GPT)	\$0.008	\$0.015	\$0.012	~\$0.05	Kharab ❌
SARVAM stack ⭐	\$0.006	\$0.005	\$0.010	~\$0.02-0.03	Best ✅ (WER 5-6%)
Sarvam + WhatsApp inbound	\$0.006	\$0.005	\$0.010	~\$0.02 (telephony free)	Best ✅
AI4Bharat self-host (at scale)	GPU only (L40S ~60 streams/\$0.72hr)			~\$0.005	Great ✅ (MIT free)

💰 **Double win:** India ke liye Sarvam se **cost aadha** (~\$0.05 → ~\$0.025) AUR **Hindi quality double**. Scale pe AI4Bharat self-host → ~\$0.005/min (10× sasta). Yaani India market me aap sabse sasta + sabse achha de sakte ho → massive margin + adoption.

6 Phased Integration Roadmap

Aapke provider-router architecture pe — mostly naye adapters, koi rewrite nahi.

Phase 1 · Week 1 — Sarvam adapters (biggest ROI) 🔥

- **3 adapters:** `provider-router/adapters/sarvam-stt.ts` (Saaras WS), `sarvam-tts.ts` (Bulbul WS), `sarvam-llm.ts` (OpenAI-compatible baseURL)
- `Provider` enum me `SARVAM` add · `pricing.ts` me Sarvam rates · voice service me Pipecat Sarvam services wire
- Routing rule: **Indic language detect** → **Sarvam auto** (translation module me script-detection already hai!)
- Result: **Hindi/regional calls turant sasti + accurate.**

Phase 2 · Week 2 — Language-aware routing + Hinglish

- Per-tenant/per-agent language preference · auto-detect (`language\_code=unknown`) · **code-mixed Hinglish mode default for India**
- Agent builder me **22-language voice picker** (Bulbul 11 + fallback) · mid-call language auto-switch

Phase 3 · Week 3–4 — Self-host tier (AI4Bharat, MIT free)

- Voice service me **IndicConformer STT + IndicF5 TTS + IndicTrans2** self-host option (GPU) → BYOK/enterprise + CodeCanyon buyers ko near-zero cost
- **Bhashini adapter** (govt/free tier) · 8kHz→16kHz upsample handle

Phase 4 · Ongoing — India go-massive features

- **State-wise agent templates** (Maharashtra→Marathi, TN→Tamil, WB→Bengali...) · WhatsApp-first (inbound free)
- UI locales 4 → 22 Indian · Gnani.ai partner tier for BFSI contact-centers

7 India ko MASSIVE banane ki strategy 🚀

- 1 **22-language coverage (Sarvam + AI4Bharat):** Har state, har regional language — koi competitor itni breadth nahi deta. Yahi moat.
- 2 **WhatsApp-first India play:** 500M+ users, inbound AI calling FREE → sabse sasta channel. India = WhatsApp country.
- 3 **Hinglish / code-mixed native:** Indians aise hi bolte hain — global models yahan fail. Sarvam+AI4Bharat first-class.
- 4 **Tier-2/3 vernacular focus:** Local clinics, shops, coaching, real-estate — apni bhasha me AI caller. Untapped huge market.
- 5 **BYOK + self-host (AI4Bharat MIT):** Cost-conscious Indian market ke liye near-zero AI cost option → adoption↑.
- 6 **Govt/PSU via Bhashini:** Sovereign, Made-in-India, subsidized → government + PSU clients (bada, sticky market).
- 7 **"Made in India / Sovereign AI" positioning:** Data-residency + Indian models = enterprise + govt trust + PR.
- 8 **State-wise reseller network:** Har state me local reseller apni bhasha me white-label bech'e → bina sales, pan-India scale.
- 9 **Vertical templates:** Loan-recovery (Hindi), clinic-booking (Marathi), Kirana-orders (Tamil) — ready-made vernacular agents.
- 10 **Price disruption:** Indian call ~\$0.02/min cost → charge ₹2-4/min (managed), still 5-10× margin. Undercut every foreign competitor.

✅ Final recommendation

Turant karo: Phase 1 — Sarvam adapters (3 din, biggest win) → Hindi/regional calls sasti + accurate. **Default India routing:** Sarvam STT+TTS+LLM, English pe Deepgram/GPT fallback. **Scale pe:** AI4Bharat self-host (~\$0.005/min, MIT). **Go-to-market:** WhatsApp-first + state-wise vernacular templates + "Sovereign Made-in-India AI" positioning + reseller network. **Result:** VocallQ = India ka sabse sasta + sabse achha + sabse broad vernacular voice-AI = massive adoption + margin.